

Tutorial 1/Module 1

DC Circuit Analysis and Theorems

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Problem 1

The charge entering a certain element is shown in Fig. 1.20. Find the current at:

- (a) $t = 1 \text{ ms}$ (b) $t = 6 \text{ ms}$ (c) $t = 10 \text{ ms}$

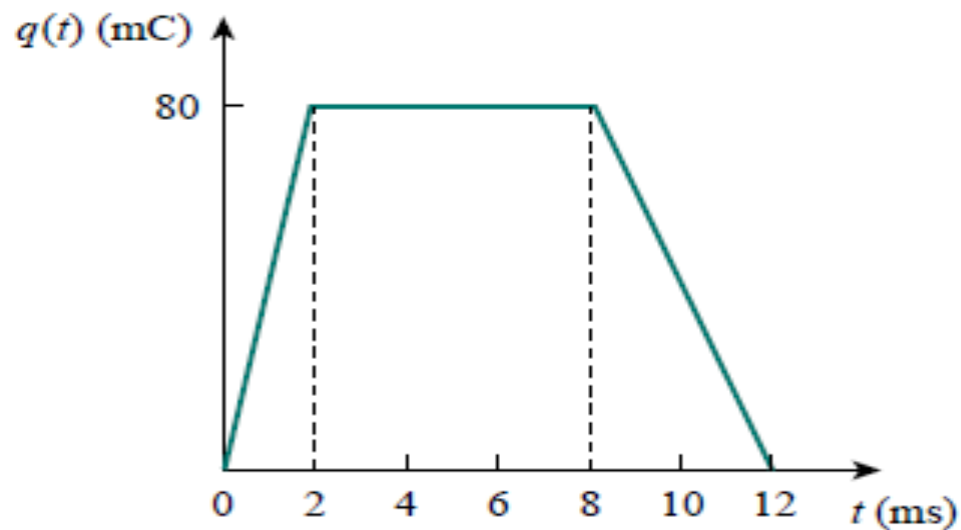


Figure 1.20

40A, 0A, -20A

Problem 2

The current through an element is shown in Fig. 1.23. Determine the total charge that passed through the element at:

- (a) $t = 1 \text{ s}$ (b) $t = 3 \text{ s}$ (c) $t = 5 \text{ s}$

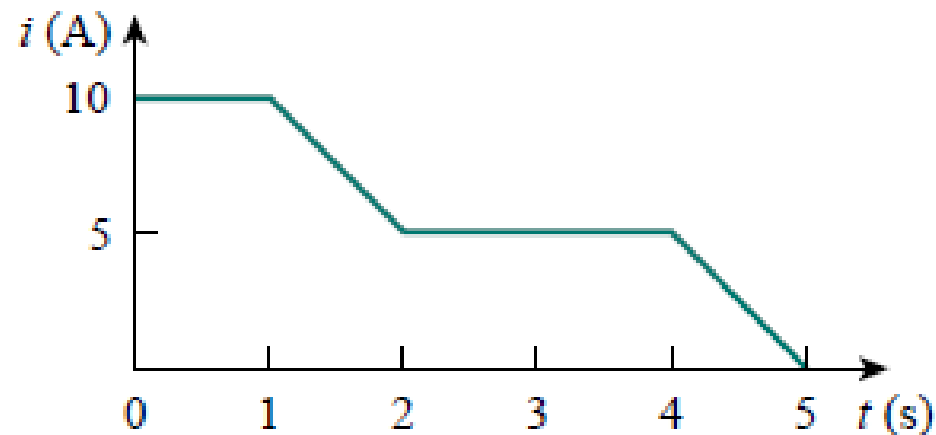


Figure 1.23

10C, 22.5C, 30C

Problem 3

Figure 1.25 shows a circuit with five elements. If $p_1 = -205 \text{ W}$, $p_2 = 60 \text{ W}$, $p_4 = 45 \text{ W}$, $p_5 = 30 \text{ W}$, calculate the power p_3 received or delivered by element 3.

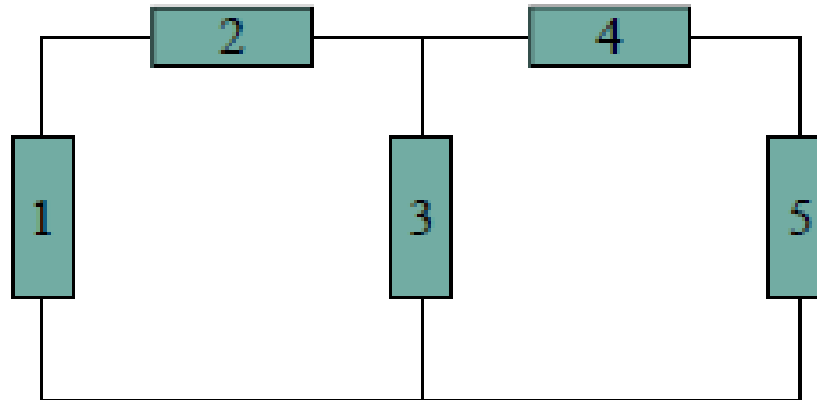


Figure 1.25

70W

Problem 4

1.15 Find the power absorbed by each of the elements in Fig. 1.26.

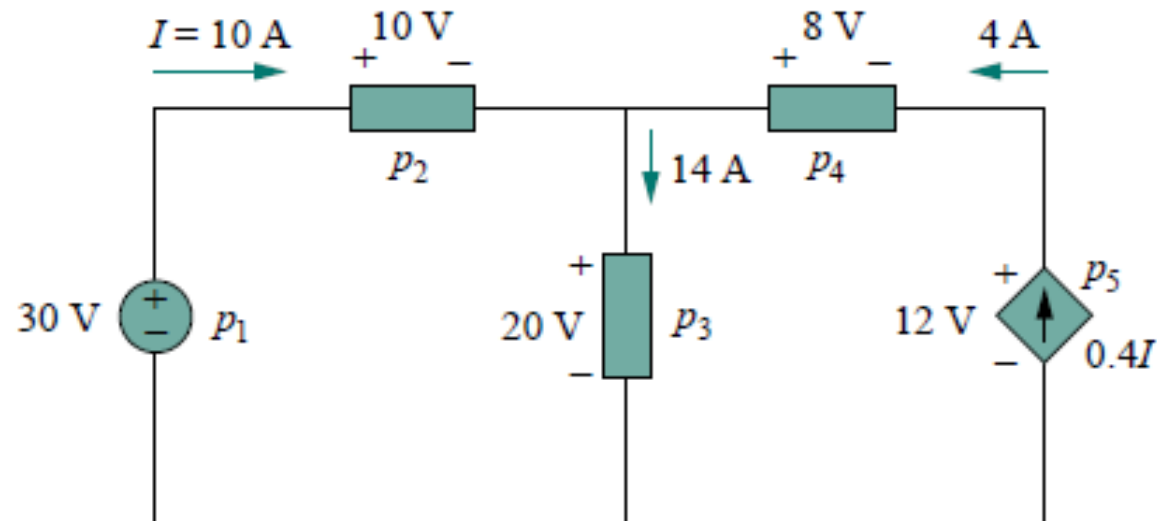


Figure 1.26

-300W, 100W, 280W, -32W, -48W

Problem 5

Find V_o in the circuit of Fig. 1.28.

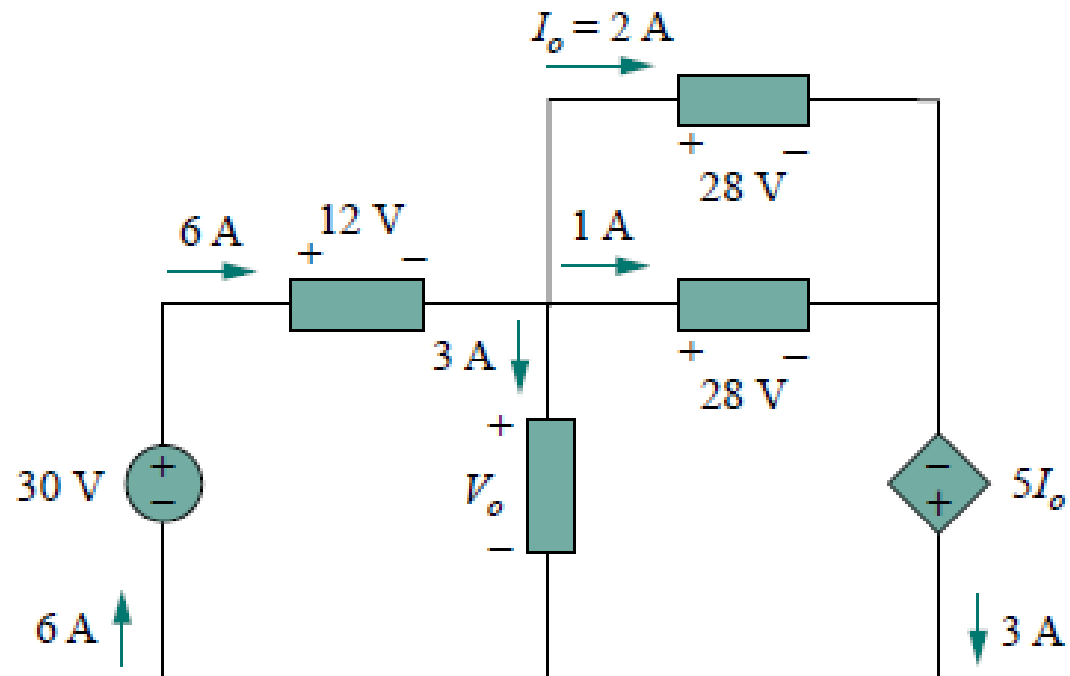
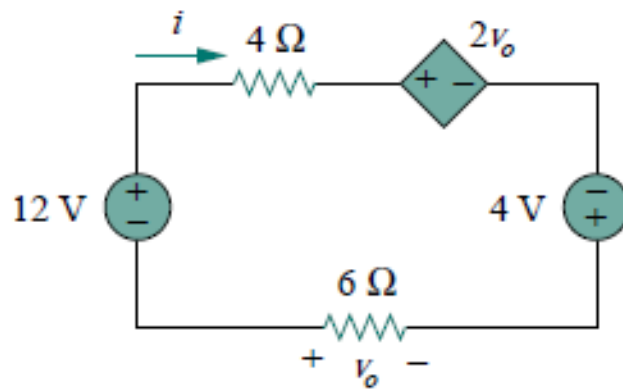


Figure 1.28

18V

Problem 6

Determine v_o and i in the circuit shown in Fig. 2.23(a).

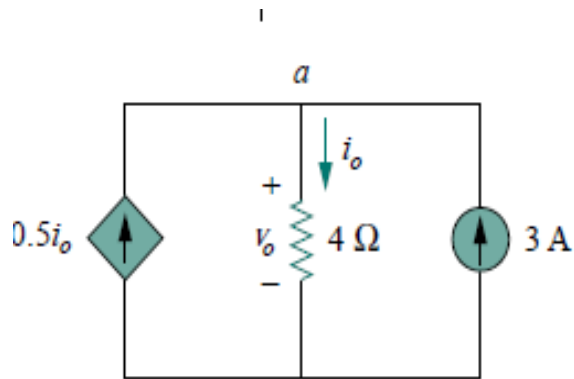


(a)

Figure 2.23

48V, -8A

Problem 7



Find current i_o and voltage v_o in the circuit shown in Fig. 2.25.

Figure 2.25

6A, 24V

Problem 8

Find the currents and voltages in the circuit shown in Fig. 2.27(a).

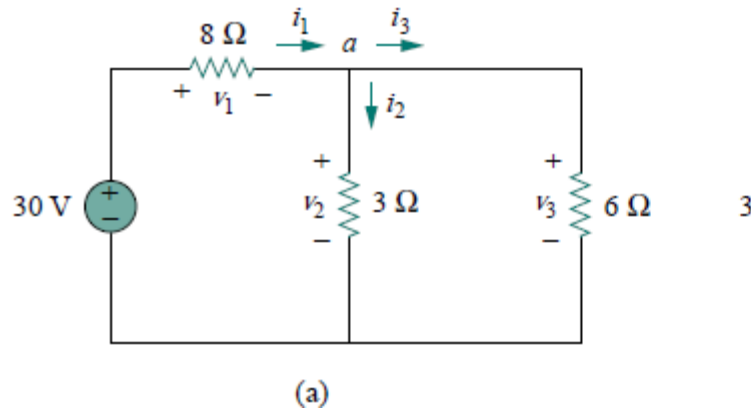


Figure 2.27

24V, 6V, 6V, 3A, 2A, 1A

Thank You